

CAP PROTECTED PHASE REVIEW

Production Marketplace Platform Capstone

THE PROJECT LEARNING LOOP

- Open only the page matching the saved attempt.
- Compare evidence, not confidence or filenames.
- Diagnose the smallest weak competency.
- Close Review before a changed retry.
- Append the retry - never overwrite attempt-01.
- No numeric percentage shortcut.

PROTECTED REMEDIATION AID - NOT A SOLUTION BOOK

Build -> prove -> break -> repair -> rerun -> defend. No solved portfolio project is supplied.

Protected route rules

Before opening

A genuine phase attempt must already exist at the exact evidence path. This Review does not authorize creating an "attempt" from its wording.

Failure sequence

saved attempt -> targeted teaching repair -> changed micro-proof -> validation -> reassessment. The original attempt and first failure remain append-only.

Reviewer standard

Look for artifact reality, independent validation, changed/failure proof, runtime truth, explanation, and unresolved critical failure. A pretty README or green placeholder test is not enough.

Protected objective controls

Course QA also has objective controls in ``CONTROL/PROTECTED_CONTROLS/CAP_expected_controls.json``. Use them only for reviewer diagnosis after attempt; they must not replace the learner's independent validation.

CAP-01 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

source_contracts.md

Failure / transfer evidence

Add/remove/change one field according to supplied compatibility rule; quarantine/fail safely rather than silently corrupt.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Starting with Airflow/dbt/Spark because they look impressive.; Writing vague goals such as "build a pipeline".; Hiding assumptions instead of recording them.. Use the recovery ladder: Rewrite the goal as: user + decision + source + output grain + freshness + failure consequence + evidence. If one field is unknown, mark it as an assumption rather than inventing certainty.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-01-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-02 - review after attempt

PASS evidence the reviewer should see

Trace every source to trusted/serving output; identify one failure boundary and replay/recovery path.

Artifact contract

architecture.md/diagram + ADRs

Failure / transfer evidence

PHASE CHECK CAP-02: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Drawing products without responsibilities.; No replay path from raw evidence.; No rejected alternative or revisit trigger.. Use the recovery ladder: Trace one source from arrival to serving. At each arrow ask: what is persisted, what can fail, what validates it, and how would I replay/recover?

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-02-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-03 - review after attempt

PASS evidence the reviewer should see

Source rows/events reconcile to raw + quarantine + deliberate duplicate handling; raw evidence remains replayable.

Artifact contract

raw/landing evidence + run_metadata.json

Failure / transfer evidence

PHASE CHECK CAP-03: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Overwriting raw input.; Calling a paginated API successful after only one page.; Dropping invalid rows without reason/evidence.. Use the recovery ladder: Separate receive/land from validate/publish. First prove exactly what arrived. Only then decide whether records are accepted, quarantined, retried, or blocked.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-03-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-04 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

tests/ + reconciliation report + quarantine evidence

Failure / transfer evidence

PHASE CHECK CAP-04: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Only testing happy-path row counts.; Using DISTINCT to hide duplication.; Letting tests exist without a publish consequence.. Use the recovery ladder: Write the invariant in plain English, define failure action, run it on a known good and known bad example, then reconcile independently.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-04-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-05 - review after attempt

PASS evidence the reviewer should see

State output grain; test key/cardinality; reconcile critical measures to an independent control query.

Artifact contract

models/SQL/dbt + lineage + row-count controls

Failure / transfer evidence

PHASE CHECK CAP-05: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Aggregating too early in staging.; Using DISTINCT to hide fan-out.; Building marts without documenting lineage.. Use the recovery ladder: Run each model/stage alone. State one-row meaning. Check keys and row counts before the join, after the join, and against an independent control.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-05-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-06 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

cap_phase_artifact + evidence note

Failure / transfer evidence

PHASE CHECK CAP-06: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Retries around non-idempotent side effects.; Downstream tasks running after partial upstream success.; Backfilling all history when only one window is damaged.. Use the recovery ladder: For each task write input, output, success condition, retry safety, and downstream dependency. Then inject one upstream failure before trusting automation.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-06-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-07 - review after attempt

PASS evidence the reviewer should see

Compare output to small trusted control result; inspect plan/partitions; justify performance choice without changing correctness.

Artifact contract

Spark job/notebook + explain plan + partition/performance evidence

Failure / transfer evidence

Supply skewed/unbalanced input and require correctness-preserving diagnosis; optimization is secondary to invariant safety.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Partitioning by high-cardinality unique id.; Optimizing before proving equivalent results.; Claiming Spark execution when only code was read.. Use the recovery ladder: Reduce to a small trusted subset, prove exact result equivalence, inspect partition counts/key distribution/plan, then change one performance variable at a time.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-07-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-08 - review after attempt

PASS evidence the reviewer should see

Map identities/connections/access/secret boundaries; prove no plaintext secret or broad unsafe access is required.

Artifact contract

Fabric deployment design + connection/access/secret boundary notes

Failure / transfer evidence

PHASE CHECK CAP-08: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Putting secrets in config/README.; Using broad admin access as a shortcut.; Calling a static design a Fabric execution.. Use the recovery ladder: Separate design evidence from runtime evidence. For every connection ask: which identity, which permission, where secret/token lives, and how access is verified/revoked?

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-08-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-09 - review after attempt

PASS evidence the reviewer should see

Define expected invariant before execution, run task, compare actual vs expected, and save one independent control check.

Artifact contract

README.md + runbook.md + evidence_index.md

Failure / transfer evidence

PHASE CHECK CAP-09: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: CI that only checks syntax.; Monitoring without response threshold/action.; Rollback plan that cannot restore data/state safely.. Use the recovery ladder: Trace change -> test -> approve -> deploy -> observe -> rollback. If any arrow has no evidence or owner, production readiness is incomplete.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-09-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-10 - review after attempt

PASS evidence the reviewer should see

Save first failing evidence before repair; identify root cause; apply smallest safe repair; fresh retry must pass all prior controls.

Artifact contract

first-failure evidence + root_cause.md + repair + fresh retry evidence

Failure / transfer evidence

MANDATORY SUPPLIED FAILURE: capture first failure, diagnose, repair, fresh retry; do not overwrite original evidence.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Deleting the failed run.; Changing multiple things at once.; Calling symptom removal a root cause.. Use the recovery ladder: Freeze failure evidence -> state expected vs actual -> isolate boundary -> test one hypothesis -> smallest repair -> changed retry -> re-run previous critical controls.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-10-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-11 - review after attempt

PASS evidence the reviewer should see

Run same input twice and prove trusted logical result is stable; then add a changed/new record and prove only intended state changes.

Artifact contract

state ledger/watermark + rerun/backfill evidence

Failure / transfer evidence

Replay the same batch/event plus one late/changed record; detect duplicate/logical-state corruption if implementation is unsafe.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Demo works only on the original machine state.; Reading README verbatim as defense.; Claiming unexecuted cloud/runtime work.. Use the recovery ladder: Use defense order: business goal -> source/grain -> architecture -> critical controls -> failure/recovery -> runtime truth -> biggest trade-off -> revisit trigger.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-11-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.

CAP-12 - review after attempt

PASS evidence the reviewer should see

Reviewer can locate every evidence artifact and challenge one design trade-off; learner explains limitation and recovery path without reading a script.

Artifact contract

defense_notes.md + reviewer checklist

Failure / transfer evidence

PHASE CHECK CAP-12: one changed-input or broken-assumption test must be run and documented before mastery.

Targeted diagnosis

If weak, name only the failed competency. Likely causes for this phase: Demo works only on the original machine state.; Reading README verbatim as defense.; Claiming unexecuted cloud/runtime work.. Use the recovery ladder: Use defense order: business goal -> source/grain -> architecture -> critical controls -> failure/recovery -> runtime truth -> biggest trade-off -> revisit trigger.

Fresh changed retry rule

Close this Review. Change one relevant input, assumption, failure condition, scale/security boundary, or source defect. Solve again without copying Review language. Re-run all previously correct critical controls and append retry evidence.

Reviewer questions

- Can I locate M18-CAP-12-EVIDENCE and the real artifact?
- What independent control could expose a plausible-but-wrong result?
- What changed/bad input was tested?
- Which runtime claim was executed vs static/not executed?
- Can the learner explain why this design/control works and what would trigger redesign?

Move on only when

L3 independent professional; no unresolved critical failure; reproducible artifact/evidence.